Summary: Vector Dimensionalities

- Total experimental conditions per colour:
  - 30 $\{V, F.r\}$ pairs
  - 5 dt_2 per $\{V, F.r\}$
  - 6 coverage levels per $\{V, F.r, dt_2\}$

$30 \times 5 \times 6 = 900$ conditions / colour

$\times$ 4 colours

$\boxed{3600}$ experimental conditions (M)

- We can separate the global experimental conditions and the per-nozzle conditions:

→ Global experimental conditions:
- Colour (4D if one-hot encoded)
- Coverage (1D)

$$\boxed{\theta^{(M)} \in \mathbb{R}^5}$$

→ Per nozzle experimental conditions:

every nozzle j belongs to one chip $h(j) \in \{1, \dots 30\}$

↳ then: * chip level parameters:

$$c_h = \begin{bmatrix} V_h \\ F.r_h \end{bmatrix} \in \mathbb{R}^2$$

* nozzle level parameters:

$$d_j = dt_{2\,j} \in \mathbb{R}$$

* combined into one input vector per nozzle:

$$u_j \begin{bmatrix} V_{h(j)} \\ F.r_{h(j)} \\ dt_{2\,j} \end{bmatrix} \in \mathbb{R}^3$$

* which gives us

$$\boxed{U^{(M)} = \mathbb{R}^{33600 \times 3}} \longrightarrow U = \begin{array}{cc} & \overbrace{\hphantom{V_{h(1)} \quad Fr_{h(1)} \quad dt2_{1}}}^{3} \\ 33600 \left\{ \right. & \begin{bmatrix} V_{h(1)} & Fr_{h(1)} & dt2_{1} \\ V_{h(2)} & Fr_{h(2)} & dt2_{2} \\ \vdots & \vdots & \vdots \\ V_{h(33600)} & Fr_{h(33600)} & dt2_{33600} \end{bmatrix} \end{array}$$

here rows are physically ordered

→ output: $\boxed{Y^{(M)} \in \mathbb{R}^{33600}}$

Combining all 3600 experimental conditions:

Input: $U \in \mathbb{R}^{3600 \times 33600 \times 3}$

Global settings: $\theta \in \mathbb{R}^{3600 \times 5}$

Outputs: $Y = \mathbb{R}^{3600 \times 33600}$

we go back to the definition:

$$f : (\theta, U) \longrightarrow Y$$

where θ : global settings $\longrightarrow \theta \in \mathbb{R}^5$

U : per-nozzle settings (spatial field) $\rightarrow U \in \mathbb{R}^{33600 \times 3}$

Y : spatial field $\rightarrow Y \in \mathbb{R}^{33600}$

- spatial index $j \in \{1, \dots, 33600\}$
- chip mapping $h(j) \in \{1, \dots, 30\}$
- input field $U_j = (V_{h(j)}, Fr_{h(j)}, dt2_j)$